

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Tiffany J. Myers, Permits Section, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: May 2, 2013

SUBJECT: Cadillac Waste Water Treatment Plant (WWTP) - Permit Review Recommendations for Toxics
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the NPDES permit application from the City of Cadillac for the Cadillac WWTP. The facility is requesting to continuously discharge a maximum of 3.2 million gallons per day (MGD) of treated municipal wastewater to the Clam River via outfall 001. The lowest monthly 95 percent exceedance, harmonic mean, and 90-day 10-year flows for Clam River upstream of outfall 001 are 2.7, 13, and 3.3 cubic feet per second (cfs), respectively.

Sources of information used for this review to develop the water quality based effluent limits (WQBELs) include the facility's NPDES permit application, the Permits Section Toxics facility file, the current NPDES permit, discharge monitoring report data, Compliance Sampling Inspections (CSI) conducted at the facility in January 2011 and September 2009, and United States Geological Survey topographical maps. Based upon the review of this information, we have the following recommendations:

- 1) The 2012 Water Quality and Pollution Control in Michigan Sections 303(d), 305(b), and 314 Integrated Report indicates the Clam River is Not Supporting the Fish Consumption designated use due to polychlorinated biphenyls (PCBs). A Total Maximum Daily Load (TMDL) for PCBs is scheduled to be developed in 2013.
- 2) The current NPDES permit includes daily minimum and maximum pH limits of 6.5 and 9.0 Standard Units (S.U.), respectively with samples collected 5X/week as a grab sample. Levels of pH in the outfall 001 final effluent since January 2008 ranged from 5.5 to 8.7 S.U., averaging 7.3 S.U. There were periodic exceedances of the daily minimum pH final effluent limitation. Based on this information, we recommend the draft permit retain the current limits and monitoring requirements for pH.
- 3) The NPDES permit reissuance application indicates ultraviolet disinfection is utilized to achieve compliance with the fecal coliform limitations instead of chlorine. Based on this information, we have no recommendations for total residual chlorine at this time.
- 4) The current NPDES permit includes a monthly average total phosphorus final effluent limit of 0.5 milligrams per liter (mg/l) (13 lbs/day) from May 1 through March 31 and a monthly average total phosphorus final effluent limit of 0.65 mg/l (17 lbs/day) from April 1 through April 30. Compliance with these limitations is 5X/week with samples collected as 24-hour composite samples. Concentrations of total phosphorus in the outfall 001 final effluent since January 2008 ranged from 0.0 to 1.78 mg/l (0.0 to 10.39 lbs/day). To protect the Clam River from excessive nutrients, we recommend the draft permit retain the current limitations and monitoring requirements for total phosphorus.

- 5) The current NPDES permit includes a level currently achievable total mercury limit of 10 nanograms per liter (ng/l) [0.00027 lbs/day] with compliance determined quarterly as a 12-month rolling average. A Pollutant Minimization Program (PMP) for total mercury was approved on May 17, 2005. Concentrations of total mercury in the outfall 001 final effluent since July 2008 ranged from Not Detected to 4.7 nanograms per liter (ng/l). These data indicate there is a reasonable potential for total mercury to be discharged at concentrations exceeding Michigan Water Quality Standards (WQS) in the outfall 001 final effluent. We recommend the draft permit include a level currently achievable total mercury final effluent limit of 2.0 ng/l (0.00005 lbs/day). Compliance with these limitations is calculated as a 12-month rolling average with samples collected quarterly as grab samples. The PMP for total mercury approved in May 2005 should be retained in the draft NPDES permit.
- 6) The current NPDES permit includes a quarterly monitoring requirement for total selenium with samples collected as 24-hour composite samples. Concentrations of total selenium in the outfall 001 final effluent since November 2008 ranged from Not Detected to 5 micrograms per liter (ug/l). Based on this data, there is not a reasonable potential for total selenium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit remove the quarterly monitoring requirement for total selenium.
- 7) The current NPDES permit includes a quarterly monitoring requirement for available cyanide with samples collected as grab samples. Concentrations of available cyanide in the outfall 001 final effluent since November 2008 ranged from Not Detected to 17 ug/l. Based on this data, there is a reasonable potential for cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit include a monthly average WQBEL for available cyanide of 5.9 ug/l (0.16 lbs/day). Compliance with these limitations should be weekly with samples collected as grab samples and analyzed using EPA Method OIA-1677 with a quantification level of 2 ug/l. Appropriate language should also be included in the draft permit to allow the opportunity for a monitoring frequency reduction from weekly to monthly after one year.
- 8) The current NPDES permit includes a quarterly monitoring requirement for total zinc with samples collected as grab samples. Concentrations of total zinc in the outfall 001 final effluent since August 2008 ranged from Not Detected to 715 ug/l. Based on this data, there is not a reasonable potential for total zinc to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit remove the quarterly monitoring requirement for total zinc.
- 9) Concentrations of total copper in the outfall 001 final effluent since August 2009 ranged from 4.0 to 15 ug/l. Based on this data, there is a reasonable potential for total copper to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit include a monthly average WQBEL for total copper of 19 ug/l (0.5 lbs/day). Compliance with these limitations should be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method. Appropriate language should also be included in the draft permit to allow the permittee to request a monitoring frequency reduction from weekly to monthly after one year.
- 10) The current NPDES permit includes final effluent limitations for chronic whole effluent toxicity (WET) of 1.1 chronic toxic units (TUc). Compliance with these limitations is currently monthly with samples collected as 24-hour composite samples and a reporting requirement for acute toxicity. Required test species is *Ceriodaphnia dubia* (*C. dubia*). The outfall 001 final effluent was never acutely toxic to *C. dubia* (all tests reported as 0.0 acute toxic units (TUa)). Chronic toxicity ranged from 0.0 to 2.4 TUc. These data indicate the outfall 001 final effluent does not have the reasonable potential to be acutely toxic. However, these data indicate the outfall 001 final effluent does have the reasonable potential to be chronically toxic. Based on this information, we recommend

the draft permit retain the current limits and monitoring requirements for WET with the exception the monitoring frequency be reduced from monthly to quarterly. In addition, fathead minnow WET testing should be required under the conditions found in Part I.A.2, Additional Monitoring Requirements of the current NPDES permit. Four fathead minnow WET tests should be conducted consistent with the dates specified in the reissued permit for the Additional Monitoring Requirements.

If you have any questions regarding these recommendations, please contact me at (517) 241-0910.

cc: Brian Jankowski, Cadillac District Supervisor, Water Resources Division